Bitcoin Market Sentiment Analysis using Hyperliquid Historical Trading Data

Objective

The objective of this project is to investigate whether Bitcoin market sentiment (Fear & Greed Index) has an impact on trader performance. Historical trading data from Hyperliquid was combined with the Fear & Greed Index to evaluate profitability, trading behavior, and market dynamics.

Dataset Description

Dataset 1

Bitcoin Fear & Greed Index

- Date
- Classification
- Value

Dataset 2

Historical Trading Data

- Account
- Coin
- Execution Price
- Size USD
- Side
- Fee
- Closed PnL
- Timestamp

Data Preparation

The following preprocessing steps were performed:

- Loaded both datasets using Pandas.
- Converted timestamps into datetime format.
- Extracted trading dates.
- Cleaned duplicate records.
- Verified missing values.
- Merged datasets using the trading date.

Exploratory Data Analysis

The following analyses were performed:

- Market Sentiment Distribution
- Profit Distribution
- Average Profit by Sentiment
- Total Profit by Sentiment
- Coin-wise Profitability
- Win Rate Analysis
- Correlation Analysis
- Daily Profit Trend

Key Findings

Market Sentiment

Extreme Greed recorded the highest average profit per trade.

Fear generated the highest cumulative trading profit due to a higher number of trades.

Extreme Fear recorded the lowest trader win rate.

Trading Performance

The overall win rate remained below 50% across all sentiment categories.

Coin @107 generated the highest cumulative profit.

BTC and ETH remained consistently profitable.

Correlation

Fee showed a strong positive relationship with trade size.

Closed PnL showed only weak correlation with other numerical variables.

Statistical Analysis

An Independent T-Test was performed to compare trader profitability under Fear and Greed market conditions.

Results

T-statistic = -0.3867

P-value = 0.6990

Since the p-value is greater than 0.05, the analysis concludes that there is no statistically significant difference in average profitability between Fear and Greed market conditions.

Business Recommendations

- Do not rely solely on market sentiment.
- Combine sentiment with technical indicators.
- Focus on proper risk management.
- Optimize position sizing.
- Monitor volatility before entering trades.

Conclusion

This analysis explored the relationship between Bitcoin market sentiment and trader performance.

Although Extreme Greed produced higher average profits, statistical testing showed that sentiment alone is not a significant predictor of trading profitability.

Future work could include leverage analysis, machine learning models, and predictive trading strategies.